MIL-P-46258C (AR)
1 June 1979
SUPERSEDING
MIL-P-46258B (MU)
20 August 1970

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, M106 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the Projectile, 8 Inch, HE, M106.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications For

STANDARDS

MILITARY

MIL-STD-453 - Inspection, Radiographic

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DRAWINGS (see 6.9)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

7548346 - Pallet for 8 Inch Projectile, Empty or Loaded
9207909 - Projectile, 8 Inch, HE, M106, for 8 Inch Howitzer

3. REQUIREMENTS

3.1 Materials.-Materials and parts shall be in accordance with applicable specifications and drawings.

3.1.1 Riser scrap shall be blended with virign explosive only in an amount such that the 1.3% maximum desensitizer wax content will not be exceeded for the melt (see 4.5.5). Clean scrap produced in regular operations may be reused provided that it complies with the requirements for the grade of high explosive specified except for form. Scrap from floor sweepings, ventilation, and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosives shall not be used (see 6.5).

3.2 Assembled projectile.-The assembled projectile shall comply with all requirements specified on Drawing (Dwg.) 9207909, and with all requirements specified in the applicable specifications.

3.3 Cavitation in high explosive charge

3.3.1 Cracks and projected cavities.-Cavities having a maximum (max.) projected length of 1/32 inch (in.) or less shall be disregarded. Cavities and cracks within the explosive charge shall not exceed the requirements specified in Table I when determined in accordance with 4.5.1.1.

3.3.3 Porous areas.-Porous areas shall be treated as cavities and shall be subject to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for

3.3.4 Cracked charges.-Not more than one transverse crack shall be permitted in Segment A and not more than a total of two transverse cracks shall be permitted in Segments A, B, and C, combined. Cracks shall be disregarded in Segment D.

3.3.5 Base separation.-There shall be no base separation in excess of fifteen thousands (.015) of an inch between the base of the shell cavity and the cast when determined in accordance with 4.5.3.

3.4 Loading of high explosive

3.4.1 Mating surfaces.-The mating surfaces of threads shall be free of caked explosives, and all threads shall be free of foreign material. If a chemical cleaner is required, acetone or methylene chloride shall be used. Care should be taken to prevent the chemical cleaner from coming into contact with the charge and becoming entrapped in the projectile.

3.4.2 Solid explosives.-Solid explosives shall not be present in the projectile before loading or be added to the shell during initial pouring or cooling.

3.4.3 Coating.-Projectile having the interior wall imperfectly coated shall not be loaded. The interior surface of the projectile shall be dry when the high explosive is poured into the projectile (see 4.4.2.1).

3.4.4 Projectile interior cavity.-The interior cavity of the projectile prior to loading shall be free from foreign matter.

3.4.5 Fuze well.-The fuze well shall be free from any foreign matter or loose explosive prior to assembly of the liner.

3.4.6 Unsolidified pool of paint.-There shall be no unsolidified pool of paint or blister in the bottom of the cavity prior to loading of the explosive.

3.4.7 Exterior surfaces.-There shall be no explosive on the exterior surface of the projectile after loading.

3.4.8 Body cavity.- The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, folds, pits and sharp edges.

3.5 Proving ground.- There shall be no premature burst or indication of metal parts failure in the gun bore or in flight before impact, when the loaded projectile is tested between 44,300 and 48,100 pounds per square inch (psi) chamber pressure.

3.6 First article inspection.- This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Exudation.- After heat cycling of TNT and Comp B filled projectiles, there shall be no exudation in the fuze well or exterior of the projectile.

3.8 Workmanship.- The requirements of workmanship are as shown on the applicable drawing, in the applicable specification and the following:

3.8.1 Cleaning.- If required the cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.8.2 Protection of the rotating band.- Care shall be exercised at all times so that the rotating band of each projectile will not be nicked or burred.

3.8.3 Painting.- Painting shall comply with the requirement of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification

3.8.4 Foreign matter.- No part or assembly shall contain dirt, grease, chips, rust corrosion or other foreign matter, prior to inspection and shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspections shall be performed on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Projectile	9207909	20

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection.-See MIL-A-48078.

TABLE II - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46258C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Projectile, 8 Inch, HE, M106				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile (Dwg. 9207909) Examination for Defects	20		3.2	4.4.2.1 to 4.4.2.5

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARRA-QA Form 2567 Feb 71 Which is Obsolete

4.4 QUALITY CONFORMANCE INSPECTION

4.4.1 Inspection lot formation.-Inspection lots shall conform with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded projectiles shall contain:

a. Projectile metal parts of one lot interfix number from one manufacturer.
b. Loaded projectile of one weight zone.
c. Fuze well liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification, and same specification revision.
d. TNT of the same type from one lot interfix number from one manufacturer, or Composition B of the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.
e. Supplementary charges from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.4.2 Examinations.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46258C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9207909
4.4.2.1	Projectile Prior to Loading				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Foreign matter in charge cavity		100%	3.8.4	Visual
2	Unsolidified pool of paint or blister in cavity		100%	3.4.6	Visual/Manual
3	Interior metal defects		100%	3.4.8	Visual
Major					
101	Interior protective coating with bare spot		0.40%	3.4.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 71 Which is Obsolete

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46258C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Projectile and Liner (Prior to Assembly of Liner)				9207909
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Depth of fuze well, maximum		0.40%	3.2	Gage
102	Foreign matter in fuze well		0.40%	3.4.5	Visual
Minor					
201	Silicone compound not 360 degrees around liner or on its base (exterior surfaces only)		100%	3.2	Visual
202	Diameter of fuze well, maximum		0.65%	3.2	Gage
203	Cavitation in base of fuze well		0.65%	3.3.2	Visual
204	Fuze well sidewall cavitation excessive		0.65%	3.3.2	Visual
205	Presence of explosive on exterior of shell body		0.65%	3.4.7	Visual
206	Evidence of poor workmanship		0.65%	3.8	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46258C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Projectile, Prior to Assembly of Lifting Plug				9207909
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Liner missing		100%	3.2	Visual
2	Supplementary charge pad or supplementary charge missing		100%	3.2	Visual
Major					
101	Concentricity of inside diameter of liner with pitch diameter of thread including perpendicularity with face of body		0.40%	3.2	Gage
102	Depth to bottom of liner, min.		0.40%	3.2	Gage
103	Supplementary charge inverted		0.40%	3.2	Visual
104	Supplementary charge not removable by hand		0.40%	3.2	Visual/Manual
Minor					
201	Liner damaged		0.65%	3.2	Visual
202	Supplementary charge damaged		0.65%	3.2	Visual
203	Foreign matter in liner cavity including explosive		0.65%	3.2	Visual
204	Explosive in thread		0.65%	3.4.1	Visual
205	Liner removable by hand		0.65%	3.2	Manual
206	Evidence of poor workmanship		0.65%	3.8	Visual
207	Spacer missing or damaged		0.65%	3.2	Visual
208	Silicone compound missing from threads		0.65%	3.2	Visual

NOTES:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46258C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Projectile Assembly				9207909
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Projectile zone weight incorrect		100%	3.2	Balance
2	Zone stenciling or punch marking incorrect		100%	3.2	Visual
Major					
101	Diameter of bourrelet, max. (applies if bourrelet has been retouched)		100%	3.2	Gage
102	Torque test failure of lifting plug		0.40%	3.2	Test
103	Lifting plug gasket missing		0.40%	3.2	Visual
104	Any weight zone punch mark inadequate		0.40%	3.2	Visual/Manual
Minor					
201	Grommet missing, damaged or improperly assembled		0.65%	3.2	Visual
202	Rotating band damaged (applies only if grommet is missing)		0.65%	3.8.2	Visual
203	Protective coating with total damaged area in excess of 1/4 square inch		0.65%	3.8.3	Visual
204	Marking misleading or unidentifiable (except for zones), including incorrect color code		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46258C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Pallet				7548346
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pallet damaged or improperly assembled		0.40%	3.2	Visual
102	DOD or NSN symbol missing or unidentifiable		0.40%	3.2	Visual
103	Projectile of different zones on the same pallet		0.40%	3.2	Visual
104	Strapping or staple missing or loose		0.40%	3.2	Manual
Minor					
201	Number of projectile assemblies incorrect		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Strapping or staple improperly engaged		0.65%	3.2	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

4.4.3 Testing

4.4.3.1 Cavitation- (see 3.3 except 3.3.2).- Cavitation defects are classified as follows:

Critical

1. Cavitation, cracks or annular rings in Segment A or B in excess of that permitted in Table I.
2. Cavitation or annular rings in Segment C in excess of that permitted in Table I.
3. Cracks in Segment C in excess of that permitted in Table I.
4. Cracks in excess of the number permitted by 3.3.4.
5. Base separation (see 4.4.3.1.1).
6. Cavitation in Segment D in excess of that permitted in Table I.

The sampling plan and procedures for this test shall be in accordance with MIL-STD-1235, using a CSP-1 plan using an AQL of 0.010 percent. The appropriate production interval or f, i values shall be as specified by the procuring activity. The test shall be performed as specified in 4.5.1, using equipment in accordance with 4.4.4.

4.4.3.1.1 Base separation - Critical defect.- Projectiles showing the largest base separation as determined by x-ray examination prior to post cycle heat treatment shall be selected for sectioning at the following rate. Select projectiles at a rate of one (1) out of five hundred (500) projectiles to be included in the post cycle heat treatment lot with a minimum of two (2) samples per lot. These projectiles are to be included in the post cycle heat treatment and sectioned for determination of base separation. Thereafter, if insufficient samples are available with base separation, a sufficient number of samples shall be randomly selected from the post cycle heat treatment lot to make up the required sample. The samples selected for sectioning must be placed in the coolest area of the post cycle igloo. Any projectile after sectioning which has base separation exceeding .015 of an inch shall be clased defective, and the entire post cycle heat treatment lot rejected.

4.4.3.2 Proving ground.- Five projectiles shall be randomly selected from the lot for testing as specified in 4.5.2. If any projectile fails to comply with the applicable requirement, the lot shall be rejected.

4.4.3.2.1 Test validity.- If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be considered invalid and a new test shall be performed with additional samples.

4.4.4 Inspection equipment.- The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.4.5 Exudation - Minor defect.- Six projectiles from the hottest area of the post cycle oven shall be used for the sample. Any projectiles with exudation shall be cause for rejection of the post cycle treatment lot.

4.5 Test methods and procedures

4.5.1 X-ray examination

4.5.1.1 Determination of projectile cavitation.- The projectiles shall be subjected to radiographic examination. The radiographic beam shall be located in a plane which is perpendicular to the longitudinal axis of the projectile and tangent to the base of its internal cavity. The outline of the explosive charge shall be clearly defined at all points. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of explosive under inspection, with the exception of specified penetrameters and radiation screening required to reduce film densities to readable levels in the thinner areas of the shell. Such screening should be free of any radiographically observable features in the projected area under inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of the radiographic technique shall be 2.0 percent and the minimum photographic density of the film used shall be no less than 1.25 at every point within the projected area of explosive, including the base, with the maximum density no greater than to permit adequate inspection over the entire area of interest, as determined in accordance with 4.5.1.3 and 4.5.1.4. All defect-like features in the area of interest on a radiographic image shall be treated as defects unless identified as extraneous

because of difficulty in evaluating the size, location, or classification of a defect, the projectile shall be rotated 90 degrees about its longitudinal axis from the original exposure and radiographed. Acceptability in all cases shall be judged on the basis of the film giving the largest projection, except in the case where a re-radiograph reveals multiple superimposed cavities, each of which individually complies with the requirements. All additional radiographs of a projectile for defect conditions will be identified on the film as retakes and stored; traceable to the original.

4.5.1.2 Radiographic equipment, operations and procedures.- Radiographic equipment, operations and procedures shall be qualified in accordance with Standard MIL-STD-453.

4.5.1.3 Check of sensitivity of radiographic technique.- The sensitivity of radiographic technique shall be determined by means of penetrameters fabricated as specified herein. The penetrameters shall be placed on the radiographic source side of a projectile so that the 2T holes are at the point shown in Figure 2. Three penetrameters for each projectile shall be used. When radiographing a number of similar projectiles the frequency of the sensitivity check shall be no less than one in each fifty radiographs, and on a rotating ring, the penetrameters shall be used at one station for each complete rotation of the ring.

The sensitivity of the radiographic technique shall be considered satisfactory when the image of the penetrameters, including the 2T holes, are perceptible in accordance with MIL-STD-453.

The penetrameters shall be made of steel having an approximate specific gravity of 7.8. The penetrameters shall be located as shown on Figure 2. For further details on construction of the penetrameter, see MIL-STD-453.

4.5.1.4 Check of photographic density of film.-The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive charge. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units. Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurement on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being

measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more thanl/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.1.5 Review of radiographic sensitivity and techniques - (see 6.8).- The sensitivity of the radiographic techniques shall be reviewed by submitting one radiograph of the standard and a copy of radiographic procedures prior to the start of initial production. These radiographs shall be taken and processed using the same techniques as will be used in production. In addition, one radiograph of each of the projectile standards shall be submitted for review once a month for the first three months of each contract and every three months thereafter.

4.5.1.6 Radiographic film.- Only a film type and vendor as listed herein has been tested and approved by ARRADCOM for use in radiographing this projectile. A substitute item shall not be used before prior testing and/or approval by ARRADCOM, Dover, New Jersey 07801.

Eastman Kodak Company
1187 Ridge Road
West Rochester, New York 14650

Film Types: AA, T, or M

E.I. DuPont de Nemours and Company
Photo Products Department
Wilmington, Delaware 19898

Film Types: NDT-55, NDT-65 or NDT-70

4.5.2 Proving ground.- This test shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample projectiles with dummy fuzes shall be fired with a Zone 9 M188 series propelling charge. The charge shall be adjusted to attain between 44,300

and 48,100 psi in a cannon with a minimum of 50 percent wear life remaining. The sample projectiles shall be conditioned and fired at 65 to 75°F. A Smear Camera shall be used to observe the projectiles in flight in order to record any possible malfunction (see 6.6). In addition, radar shall be used to observe the projectile in flight in order to record any possible malfunction.

4.5.3 Determination of base separation.- After heat treatment, the samples from each post cycle heat treatment lot shall be cooled to ambient and sectioned in a manner to expose the interface between the cast and the interior base of the projectile at the approximate center of the base. The projectile shall be positioned nose down or the cast shall be moved as far towards the nose with finger pressure as possible before measuring for base separation between the shell cavity and cast.

4.5.4 Exudation.- The exterior of the projectile shall be visually inspected for exudation. If no exudate is found, remove the lifting plug and supplementary charge and inspect for exudation visually.

4.5.5 Determination of desensitizer content.- Riser scrap shall be batched in amounts appropriate to the production rate but not in excess of a 1 shift accumulation. For each batch of scrap, a sample of 4 randomly selected risers shall be crushed and blended. A 5 gram sample shall then be selected for determination of desensitizer content in accordance with MIL-C-401E. The amount of riser that can be added to virgin material shall then be calculated based on its desensitizer content, the desensitizer content of the virgin explosive as specified on the explosive data sheet and the total kettle weight.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level A.- The projectile shall be packed in accordance with Dwg. 7548346.

5.2 Marking.- Marking shall be in accordance with Dwg. 7548346.

6. NOTES

6.1 Intended use.- The components and assemblies covered by this specification are intended for use on the M106 Projectile.

6.2 Ordering data.- See MIL-A-48078 (PA)

6.3 Submission of designs for approval - See 6.2.3 of MIL-A-48078.-Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Photographic densities.-"American Standards for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, has been satisfactory as a source of informatition on film densities.

6.5 Clean explosive scrap.-Clean explosive scrap is defined as scrap in kettles, pails, risers and loading machines used in the pouring and cooling operations and scrap formed during the drilling operation.

6.6 Proving ground.-A copy of all proving ground data should be submitted to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q and DRDAR-LCU-EP, Dover, New Jersey 07801.

6.7 Visual examination qualification.-When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.8 Radiographic procedures.-Radiographic procedures standards and sample radiographs shall be submitted to the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801.

6.9 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.10 Changes from previous issue.-Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project No.: 1320-A41

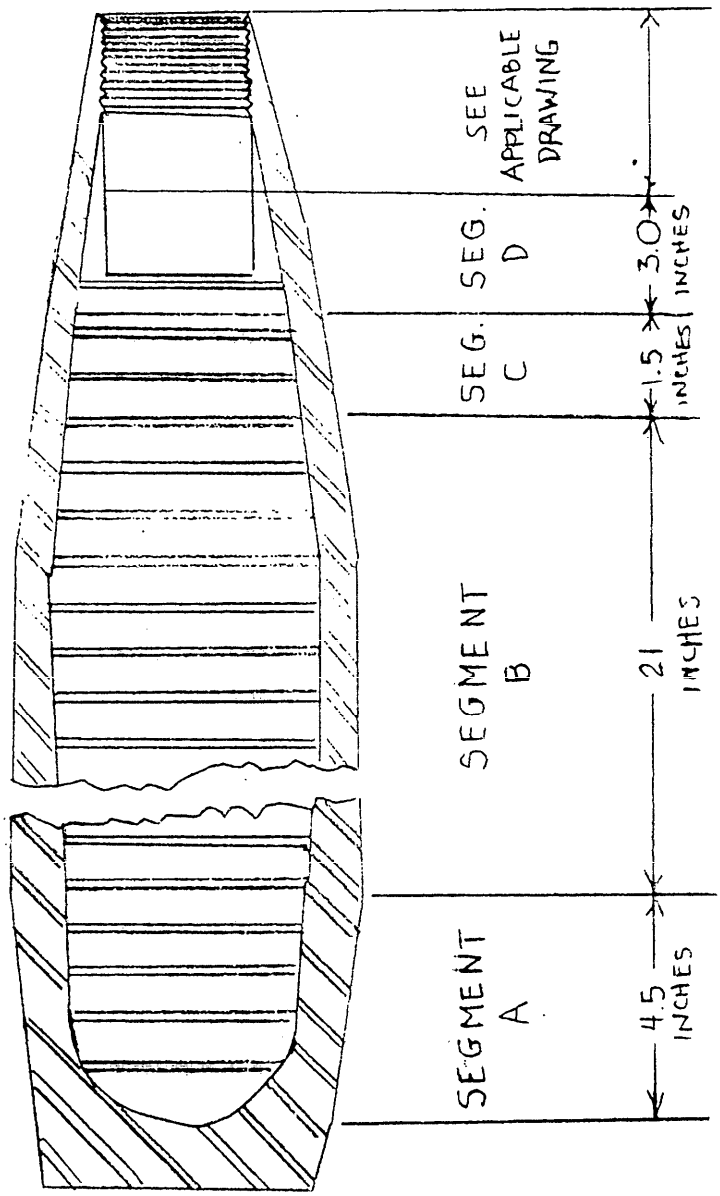
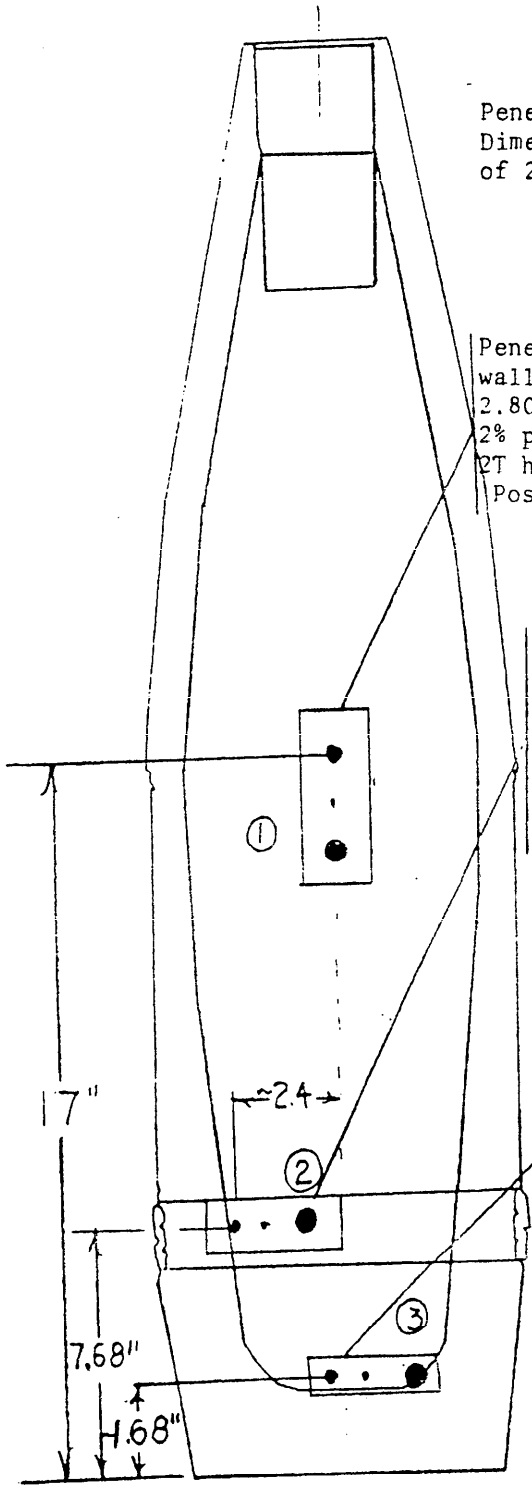

SEE APPLICABLE DRAWING
SEG. D
3.0 INCHES
SEG. C
1.5 INCHES
SEGMENT B
21 INCHES
SEGMENT A
4.5 INCHES

FIGURE 1

PROJECTILE, 8 INCH, HE, M106
RADIOGRAPHIC PENETRAMETER REQUIREMENTS



Penetrameters Constructed Per MIL-STD-453.
Dimensions shown are for location of center of 2T holes.

Penetrameter for 2.8" steel 1.50" total steel wall plus 6.468" explosive.
2.80" equivalent thickness
2% penetrameter thickness = .060"
2T hole diameter = .120"
Position 2T hole on central axis of projectile.

Penetrameter for 4.6" steel
4.25" total steel wall plus 1.328" explosive at .080" from tangent point.
4.60" equivalent thickness
2% penetrameter thickness = .080"
2T hole diameter = .160"
Position 2T hole so as tangent to the cavity wall.

Penetrameter for 5.0" steel
4.48" total steel wall plus 3.125" explosive.
5.0" equivalent thickness
2% penetrameter thickness = .090"
2T hole diameter = .180"
Position 2T hole so as tangent to the cavity base and on the central axis of projectile.

FIGURE 2

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-46258 Projectile, 8Inch, HE, M106 LAP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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